

PRODUCTION SCOPE & MILESTONE DELIVERY SCHEDULE

Live Cockpit: https://ledgerflow-core.vercel.app

MILESTONE	ARCHITECTURE & PRODUCTION ENGINEERING DELIVERABLES	TIMELINE	SHARE	ALLOCATION
Phase 0	Interactive Payments & General Ledger Cockpit (Live Prototype) Living demo: NACHA 94-character batch generator, double-entry balance verification, UUIDv4 idempotency replay interceptor, and PostgreSQL EXPLAIN plan benchmark.	Live Now	Included	\$0.00 (Live)
Milestone 1	Codebase Audit, Ingestion Schemas & Automated Testing Harness Audit current React/Node TypeScript codebases, configure strict Zod request validation, integrate Vitest/Jest integration test suites, and establish automated GitHub Actions CI/CD staging pipeline.	Week 1	20%	25 Hrs (\$2,000)
Milestone 2	ACH Payment Engine & NACHA 94-Character Batch Compilation Build specification-compliant NACHA generator handling File/Batch Headers, Entry Details (Codes 27/22, PPD/CDD/WEB), and File Controls with automated Federal Reserve routing checksum verification.	Week 2	20%	25 Hrs (\$2,000)
Milestone 3	Double-Entry Ledger Bookkeeping & Distributed Idempotency Locks Implement multi-entity general ledger ensuring atomic Sum(Debits) === Sum(Credits). Wire Redis SET NX EX 86400 locks to intercept network retries and completely eliminate duplicate settlement debits.	Week 3	20%	25 Hrs (\$2,000)
Milestone 4	PostgreSQL 16 Slow Query Tuning, Covering Indexes & Zero Downtime Profile multi-million row transaction tables with EXPLAIN (ANALYZE, BUFFERS). Deploy composite B-Trees with covering INCLUDE clauses, cutting query execution from 142ms down to 1.4ms without table locks.	Week 4	20%	25 Hrs (\$2,000)
Milestone 5	AWS ECS Fargate & SQS Worker Fleet Architecture & Handover Deploy containerized payment workers on AWS ECS Fargate with SQS FIFO queues, auto-scaling task definitions, exponential backoff retries with full jitter, and DLQ poison-pill isolation runbooks.	Week 5	20%	25 Hrs (\$2,000)
TOTAL SENIOR FULL-STACK ENGINEERING ENGAGEMENT (LONG-TERM / 25–35 HRS/WK)		Ongoing	100%	\$80.00/hr Capped

MILESTONE DELIVERY CADENCE

ARCHITECTURE GUARDRAILS & RELIABILITY INVARIANTS

Phase 0: Interactive Architecture Cockpit (Delivered)	\$0.00 (Live Ahead of Bid)	✓ Zero duplicate debits via UUIDv4 idempotency lock (SET key NX EX 86400).
W1: Codebase Discovery & Vitest CI/CD Setup	25 Hrs (Net Day 7)	✓ Strict double-entry accounting invariant (Sum Debits === Sum Credits).
W2: ACH Engine & NACHA 94-Char Batching	25 Hrs (Net Day 14)	✓ NACHA Rule 5.1 compliant 94-char fixed-width line compilation.
W3: Double-Entry Ledger & Idempotency Locks	25 Hrs (Net Day 21)	✓ PostgreSQL 16 composite covering B-Trees for sub-2ms balance queries.
W4: PostgreSQL EXPLAIN Tuning & Indexes	25 Hrs (Net Day 28)	✓ AWS ECS Fargate & SQS FIFO queues with DLQ poison-pill isolation.
W5: AWS ECS Fargate & SQS Worker Fleet	25 Hrs (Net Day 35)	✓ Claude Code AI workflows accelerating delivery with verified typing.

CALIBRATED RATE \$80.00/hr USD for 25–35 hrs/wk. Clean hourly billing aligned with profile rate and historical senior tech hires; zero rate hike clauses.	AWS INFRASTRUCTURE Self-hosted Aurora + Fargate costs ~\$98/mo, saving \$195K+/yr vs 20¢/tx vendor markups from Modern Treasury.	CODE OWNERSHIP 100% intellectual property ownership. Clean GitHub PRs, automated Vitest coverage, and comprehensive architecture docs.	EST COLLABORATION Working dedicated EST business hours in close sync with Alexander; responsive, autonomous, zero micromanagement required.
---	--	--	---

FORMAL ENGAGEMENT ACCEPTANCE & ENGINEERING AUTHORIZATION		CONTRACT REF: UPWORK-ALEX-LEDGERFLOW-2026	
LEAD SYSTEMS ARCHITECT (BARAKAHSOFT LLC)		CLIENT ENGINEERING SPONSOR (ALEXANDER)	
Shakil Ahmed		Accepted via Upwork Hourly Contract	
March 2026		Pending	
AUTHORIZED SIGNATURE • SHAKIL AHMED, PRINCIPAL SYSTEMS ARCHITECT		ALEXANDER • SENIOR FULL-STACK ENGINEER HIRING LEAD	